Sécurité des microservices avec Docker

Objectifs :

1. Mettre à jour le fichier docker-compose.yml
2. Créer le client dept-app et les rôles dans keycloack
3. Tester l'accès aux api avec le rôle ADMIN
4. Rendre les données de Keycloak permenantes

Mettre à jour le fichier docker-compose.yml

Ajouter le service keycloak au fichier docker-compose.yml

```
keycloak:
  image: quay.io/keycloak/keycloak:26.3.4
  container_name: keycloak
  command: start-dev
  environment:
    - KC_BOOTSTRAP_ADMIN_USERNAME=admin
    - KC_BOOTSTRAP_ADMIN_PASSWORD=admin
  ports:
    - "8090:8080"
```

- Modifier le service gateway comme suit :

```
environment:
  - DISCOVERY_SERVICE_URL=http://eurekaserver:8761/eureka
  - SPRING_SECURITY_OAUTH2_RESOURCESERVER_JWT_JWK_SET_URI=http://
keycloak:8080/realms/master/protocol/openid-connect/certs

depends_on:
  eurekaserver:
    condition: service_healthy
  keycloak:
    condition: service_started
```

- Afin qu'ils ne puissent être accessible, enlevez la propriété *ports* pour tous les services sauf keycoack et gatewayserver
- Afin d'alléger l'exécution, supprimer le service departments-ms2

```
services:
  eurekaserver:
    build: ./eurekaserver
    container_name: eurekaserver
#    ports:
#      - "8761:8761"
    healthcheck:
      test: "curl --fail --silent localhost:8761/actuator/health | grep UP
|| exit 1"
      interval: 10s

  configserver:
    build: ./configserver
    container_name: configserver
#    ports:
#      - "9999:9999"
    healthcheck:
      test: "curl --fail --silent localhost:9999/actuator/health | grep UP
|| exit 1"
      interval: 10s

  departments-ms:
    build: ./department-microservice
    container_name: departments-ms
    environment:
      - DISCOVERY_SERVICE_URL=http://eurekaserver:8761/eureka
      - CONFIG_SERVICE_URL=http://configserver:9999
    healthcheck:
      test: "curl --fail --silent localhost:8080/actuator/health | grep UP
|| exit 1"
    depends_on:
      eurekaserver:
        condition: service_healthy
      configserver:
        condition: service_healthy

  keycloak:
    image: quay.io/keycloak/keycloak:26.3.4
    container_name: keycloak
    command: start-dev
    environment:
      - KC_BOOTSTRAP_ADMIN_USERNAME=admin
      - KC_BOOTSTRAP_ADMIN_PASSWORD=admin
    ports:
      - "8090:8080"

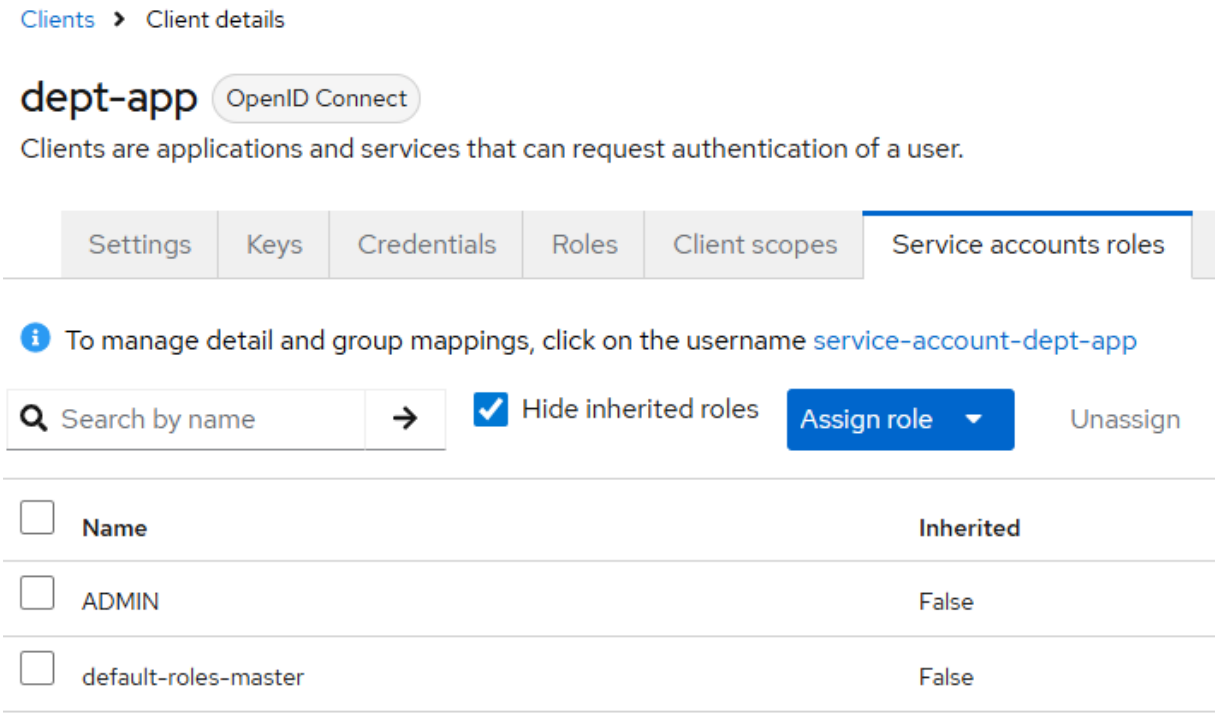
  gatewayserver:
    build: ./gatewayserver
    container_name: gatewayserver
    ports:
      - "8888:8888"
    environment:
      - DISCOVERY_SERVICE_URL=http://eurekaserver:8761/eureka
      - SPRING_SECURITY_OAUTH2_RESOURCESERVER_JWT_JWK_SET_URI=http://key-
cloak:8080/realms/master/protocol/openid-connect/certs
    healthcheck:
      test: "curl --fail --silent localhost:8888/actuator/health | grep UP
|| exit 1"
    depends_on:
```

```
      eurekaserver:
        condition: service_healthy
      keycloak:
        condition: service_started


  departments-ms2:
    build: ./department-microservice
    container_name: departments-ms2
    environment:
      - DISCOVERY_SERVICE_URL=http://eurekaserver:8761/eureka
      - CONFIG_SERVICE_URL=http://configserver:9999
    healthcheck:
      test: "curl --fail --silent localhost:8084/actuator/health | grep UP
|| exit 1"
    depends_on:
      eurekaserver:
        condition: service_healthy
      configserver:
        condition: service_healthy

  teacher-ms:
    build: ./teacher
    container_name: teacher-ms
    environment:
      - DISCOVERY_SERVICE_URL=http://eurekaserver:8761/eureka
      - CONFIG_SERVICE_URL=http://configserver:9999
    healthcheck:
      test: "curl --fail --silent localhost:8081/actuator/health | grep UP
|| exit 1"
    depends_on:
      eurekaserver:
        condition: service_healthy
      configserver:
        condition: service_healthy
      departments-ms:
        condition: service_healthy
```

Créer le client dept-app et les rôles dans keycloack



Tester avec Postman

Rendre les données de Keycloak permenantes

```
postgres:
  image: postgres:16
  container_name: kc-postgres-dev
  environment:
    POSTGRES_DB: keycloak
    POSTGRES_USER: keycloak
    POSTGRES_PASSWORD: keycloak
  volumes:
    - kc_pg_data:/var/lib/postgresql/data
  restart: unless-stopped

keycloak:
  image: quay.io/keycloak/keycloak:26.3.4
  container_name: keycloak
  command: start-dev
```

```
  environment:
    - KC_DB=postgres
    - KC_DB_URL=jdbc:postgresql://postgres:5432/keycloak
    - KC_DB_USERNAME=keycloak
    - KC_DB_PASSWORD=keycloak
    - KC_BOOTSTRAP_ADMIN_USERNAME=admin
    - KC_BOOTSTRAP_ADMIN_PASSWORD=admin
  ports:
    - "8090:8080"
  depends_on:
    - postgres

volumes:
  kc_pg_data:
```